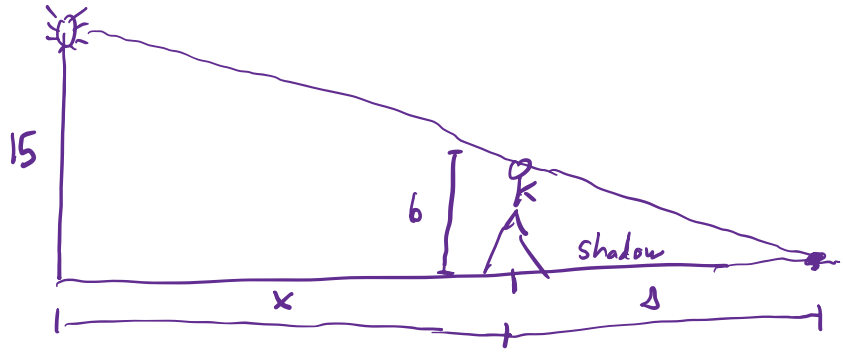
Problem 4. WORKED IN CLASS

A street light sits on top of a 15 ft pole. A man 6 ft tall walks away from the pole at 5 ft/s. How fast is the **tip of his shadow** moving when he is 40 ft from the pole?

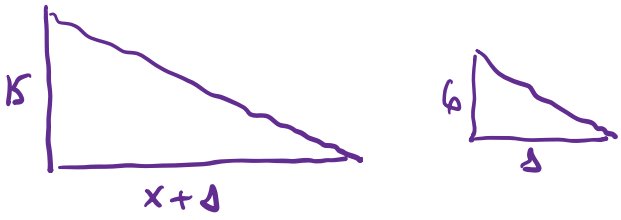
1. Draw + Label
2. What & when
3. Equation
4. Differentiate w.r.t. t
5. Sub + Solve

(1)



(2) $\frac{dx}{dt} = 5$ what we want is $\frac{d}{dt}(x+s)$ when $x = 40$

(3)



$$\frac{s}{6} = \frac{x+s}{15} \rightarrow 15s = 6(x+s)$$

$$15s = 6x + 6s$$

$$\boxed{9s = 6x}$$

(4) $\frac{d}{dt} 9s = \frac{d}{dt} 6x \rightarrow 9\frac{ds}{dt} = 6\frac{dx}{dt}$

$$9\frac{ds}{dt} = 6 \cdot 5 \rightarrow 9\frac{ds}{dt} = 30 \rightarrow \frac{ds}{dt} = \frac{30}{9} = \frac{\cancel{3}\cdot 10}{\cancel{3}\cdot 3} = \frac{10}{3}\ \frac{ft}{sec}$$

$$\frac{d}{dt}(x+s) = \frac{dx}{dt} + \frac{ds}{dt} = \frac{3}{3}5 + \frac{10}{3} = \frac{15}{3} + \frac{10}{3} = \frac{25}{3}\ ft/sec \approx 8.33\ ft/sec$$